ARTIN'S THEOREM: A FLAT COVER OF A REPRESENTABLE STACK MAKES THE TARGET ALGEBRAIC

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span; proof complete as written.

Allowed standard prerequisites (editorial).

- Diagonal representability. lemma-flat-finite-presentation-surjective-diagonal (same chapter, named). Handles the diagonal of Y.
- Algebraicity of Hilbert stacks. lemma-hilbert-stack-relative-space (same chapter, named). Algebraicity of the coproduct of H_d.
- LCI substack and its properties. lemma-lci-locus-stack-in-groupoids, lemma-limit-preserving, lemma-lci-formally-smooth, lemma-lci-surjective (same chapter, named). The lci locus is algebraic, and limit-preserving/formally-smooth/surjective on objects.
- Property implication and lifting. algebraic-lemma-representable-transformations-property-implication, spaces-more-morphisms-lemma-smooth-formally-smooth (cross-chapter, named). Transfer object-level properties to morphism-level and conclude smoothness.

Source reference. Original reference 05UN: focused-seeds/geometry/S9-crit-bootstrap/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 045F: focused-seeds/geometry/S9-crit-bootstrap/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 06D2: sources/stacks-criteria/pinned-original-source.tex, source lines 939–939. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 06CW: sources/stacks-criteria/pinned-original-source.tex, source lines 628–628. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:criteria.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 06D6: sources/stacks-criteria/pinned-original-source.tex, source lines 1077–1077. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 05XW: sources/stacks-criteria/pinned-original-source.tex, source lines 412–412. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 06CI: sources/stacks-criteria/pinned-original-source.tex, source lines 2616–2616. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 06D9: sources/stacks-criteria/pinned-original-source.tex, source lines 2905–2905. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 06CL: sources/stacks-criteria/pinned-original-source.tex, source lines 2785–2785. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 06DA: sources/stacks-criteria/pinned-original-source.tex, source lines 2953–2953. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 06CH: sources/stacks-criteria/pinned-original-source.tex, source lines 2214–2214. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 06CY: sources/stacks-criteria/pinned-original-source.tex, source lines 729–729. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 06D3: sources/stacks-criteria/pinned-original-source.tex, source lines 977–977. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 06CX: sources/stacks-criteria/pinned-original-source.tex, source lines 661–661. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 06D7: sources/stacks-criteria/pinned-original-source.tex, source lines 1096–1096. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04AM: focused-seeds/geometry/S9-crit-bootstrap/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Theorem 0.1. *Let S be a scheme. Let $F : \mathcal{X} \to \mathcal{Y}$ be a 1-morphism of stacks in groupoids over $(Sch/S)_{fppf}$. If*

1. *$\mathcal{X}$ is representable by an algebraic space, and*
2. *F is representable by algebraic spaces, surjective, flat and locally of finite presentation,*

then $\mathcal{Y}$ is an algebraic stack.

Proof. By Lemma 05XW we see that the diagonal of $\mathcal{Y}$ is representable by algebraic spaces. Hence we only need to verify the existence of a 1-morphism $f : \mathcal{V} \to \mathcal{Y}$ of stacks in groupoids over $(Sch/S)_{fppf}$ with $\mathcal{V}$ representable and f surjective and smooth. By Lemma 06CI we know that

$$\coprod_{d \geq 1} \mathcal{H}_d(\mathcal{X}/\mathcal{Y})$$

is an algebraic stack. It follows from Lemma 06CL and Algebraic Stacks, Lemma 05UN that

$$\coprod_{d \geq 1} \mathcal{H}_{d,lci}(\mathcal{X}/\mathcal{Y})$$

is an algebraic stack as well. Choose a representable stack in groupoids $\mathcal{V}$ over $(Sch/S)_{fppf}$ and a surjective and smooth 1-morphism

$$\mathcal{V} \longrightarrow \coprod_{d \geq 1} \mathcal{H}_{d,lci}(\mathcal{X}/\mathcal{Y}).$$

We claim that the composition

$$\mathcal{V} \longrightarrow \coprod_{d \geq 1} \mathcal{H}_{d,lci}(\mathcal{X}/\mathcal{Y}) \longrightarrow \mathcal{Y}$$

is smooth and surjective which finishes the proof of the theorem. In fact, the smoothness will be a consequence of Lemmas 06CH and 06D9 and the surjectivity a consequence of Lemma 06DA. We spell out the details in the following paragraph.

By construction $\mathcal{V} \to \coprod_{d \geq 1} \mathcal{H}_{d,lci}(\mathcal{X}/\mathcal{Y})$ is representable by algebraic spaces, surjective, and smooth (and hence also locally of finite presentation and formally smooth by the general principle Algebraic Stacks, Lemma 045F and More on Morphisms of Spaces, Lemma 04AM). Applying Lemmas 06CX, 06D3, and 06D7 we see that

$\mathcal{V} \to \coprod_{d \geq 1} \mathcal{H}_{d,lci}(\mathcal{X}/\mathcal{Y})$ is limit preserving on objects, formally smooth on objects, and surjective on objects. The 1-morphism $\coprod_{d \geq 1} \mathcal{H}_{d,lci}(\mathcal{X}/\mathcal{Y}) \to \mathcal{Y}$ is

(1) limit preserving on objects: this is Lemma 06CH for $\mathcal{H}_d(\mathcal{X}/\mathcal{Y}) \to \mathcal{Y}$ and we combine it with Lemmas 06CL, 06CY, and 06CW to get it for $\mathcal{H}_{d,lci}(\mathcal{X}/\mathcal{Y}) \to \mathcal{Y}$,
(2) formally smooth on objects by Lemma 06D9, and
(3) surjective on objects by Lemma 06DA.

Using Lemmas 06CW, 06D2, and 06D6 we conclude that the composition $\mathcal{V} \to \mathcal{Y}$ is limit preserving on objects, formally smooth on objects, and surjective on objects. Using Lemmas 06CX, 06D3, and 06D7 we see that $\mathcal{V} \to \mathcal{Y}$ is locally of finite presentation, formally smooth, and surjective. Finally, using (via the general principle Algebraic Stacks, Lemma 045F) the infinitesimal lifting criterion (More on Morphisms of Spaces, Lemma 04AM) we see that $\mathcal{V} \to \mathcal{Y}$ is smooth and we win. $\square$